Siva Motors AI

Intelligent Service Advisor System Architecture & Implementation

Project Overview

Transforming Automotive Support with Generative AI

Core Technology Stack

Backend Engine

Built on **Flask**, the system serves as a robust middleware connecting the frontend chat interface with AI models and data storage solutions.

Includes UUID-based session management to track unique customer interactions seamlessly.

Intelligence Layer

Powered by **Google Gemini Pro**. The system leverages LLM capabilities for both natural conversation and structured lead extraction.

Integrates `google.generativeai` for real-time diagnostic reasoning.

Gemini Pro Integration

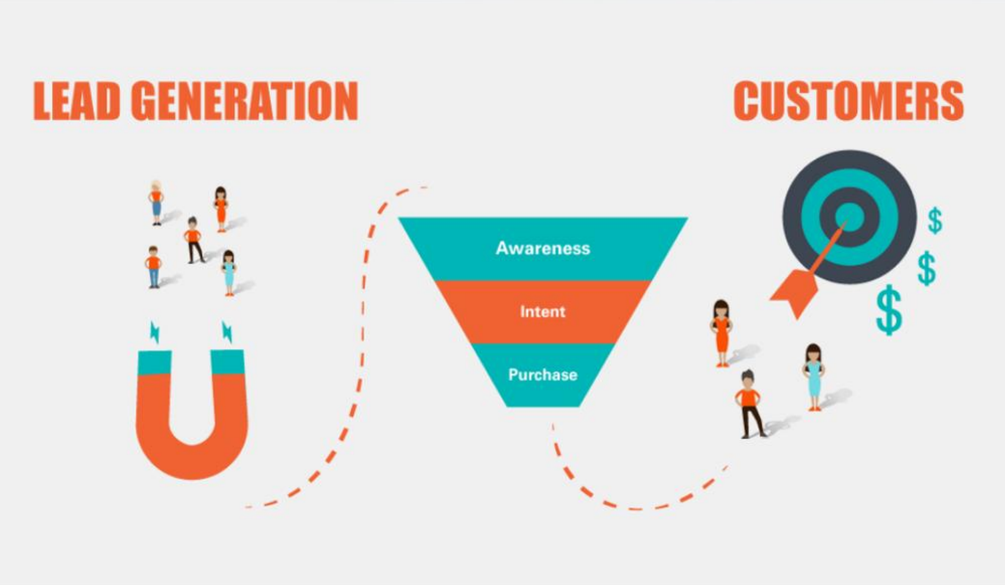
The system utilizes `gemini-pro` for high-fidelity response generation. By using a specialized **System Prompt**, Gemini adopts the persona of a professional human service advisor.

- **Natural Language Processing:** Understands complex vehicle symptoms.
- **Multi-turn Conversations:** Maintains context throughout the chat session.
- **Context-Aware:** Grounded in Siva Motors' specific service knowledge.

Automated Lead Extraction

The `extract_lead` function identifies high-intent customers by monitoring chat transcripts for contact details.

- **Validation Rule:** Triggers only when both Name and Phone are provided.
- **Output Format:** Enforces a strict JSON structure for database consistency.
- **Automated Routing:** Instantly pushes verified leads to the sales dashboard.



Cloud Data Persistence

Lead Management

Verified contact details are appended to `LEAD_SHEET_ID` via the gspread API.

Transcript Logs

Complete chat histories are logged in the `TRANSCRIPT_SHEET` for quality analysis.

Oauth2 Security

Service Account Credentials ensure secure, automated access to Google Workspace.

The "Human-Like" Persona

"It could be a battery issue or starter problem. Are you hearing any clicking sound when you try to start it?"

Siva Motors AI Persona Guidelines

Designed for brevity (2-4 lines), empathy, and proactive follow-up questions.

Response Optimization

~0s

Keyword Routing Delay

Smart Logic Routing

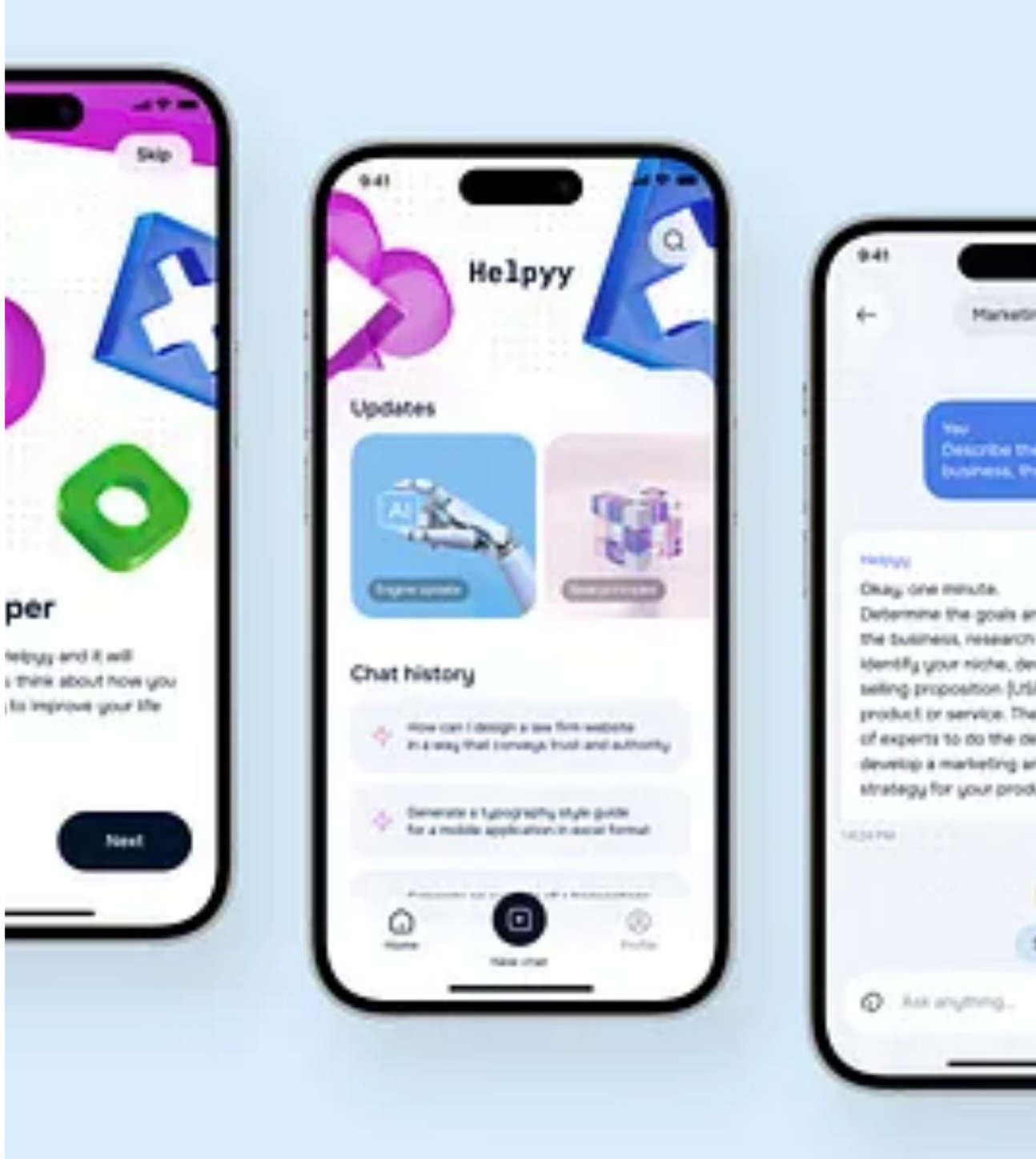
To reduce LLM latency and API costs, the system uses **Smart Routing** for high-frequency queries like Pricing and Service schedules.

Keywords trigger direct database functions (`get_price_details`) instead of generating new AI responses.

Front-End Experience

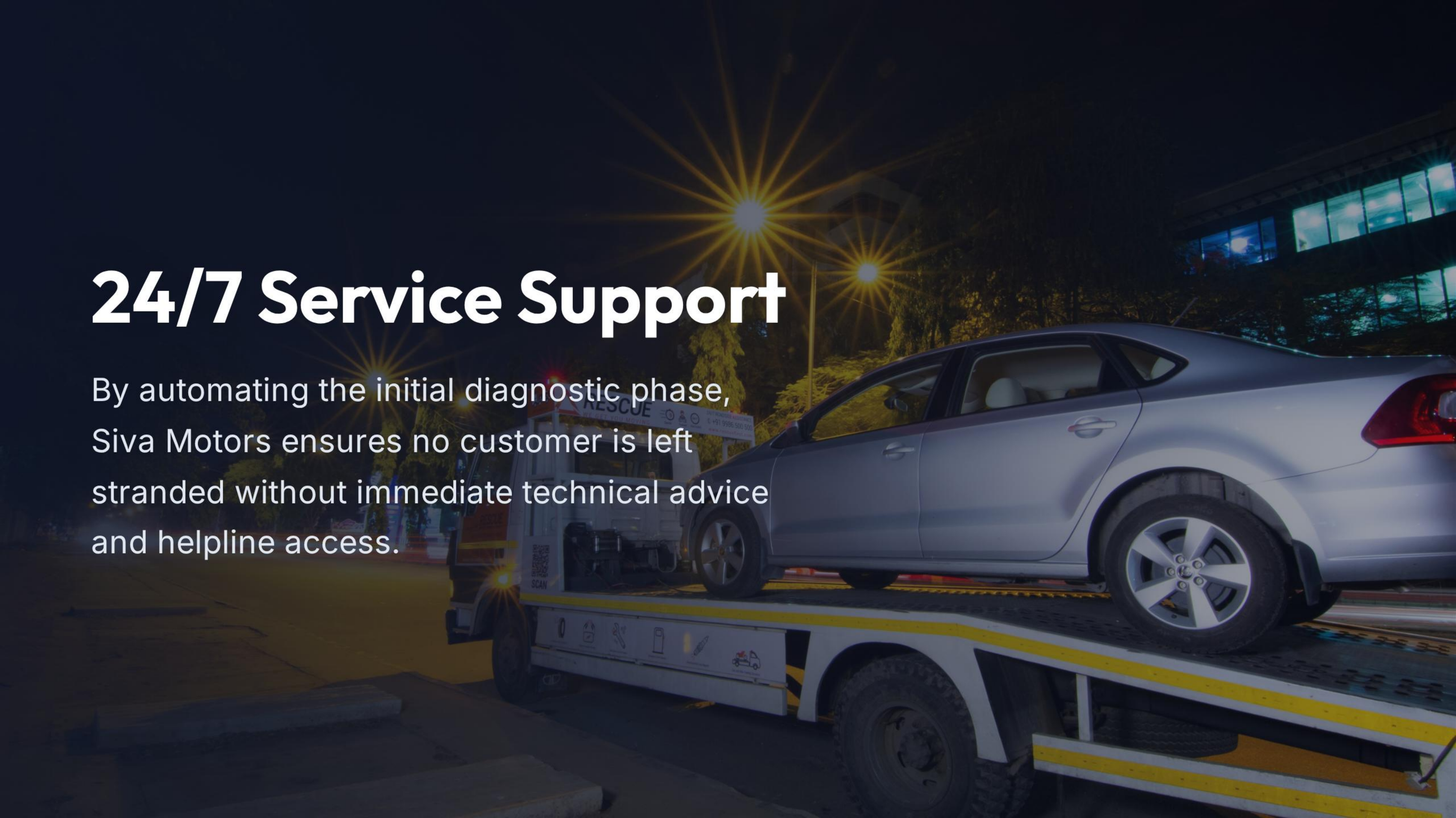
The web interface features a clean, mobile-responsive chat window designed for immediate vehicle diagnostics.

- Real-time interaction with Flask API.
- Distinct AI and User chat bubbles.
- Clear "Online" status indicators.
- Integrated Helpline support links.



Routing Business Logic

Keyword Group	Triggered Function	Outcome
"Price", "Cost"	`functions.get_price_details()`	Instant static pricing list
"Service", "Repair"	`functions.get_service_details()`	Maintenance schedule retrieval
"Help", "Emergency"	Helpline Constant	Roadside Assistance +91 98765 43210
Symptoms (Noise, Smoke)	`chat.send_message()`	AI-driven diagnostic chat



24/7 Service Support

By automating the initial diagnostic phase, Siva Motors ensures no customer is left stranded without immediate technical advice and helpline access.

Questions?

Thank you for exploring the Siva Motors AI Integration.

System Architect: AI Service Implementation

Image Sources

https://png.pngtree.com/png-clipart/20250623/original/pngtree-glowing-blue-digital-brain-with-intricate-circuit-patterns-a-high-resolution-png-image_21215423.png

Source: pngtree.com

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Source: www.getrescued.in